

PHOTODISSOCIATION REGIONS. II. A MODEL FOR THE ORION PHOTODISSOCIATION REGION

A. G. G. M. TIELENS AND DAVID HOLLENBACH

NASA/Ames Research Center, Moffett Field, California

Received 1984 July 13 accepted 1984 October 31

ABSTRACT

We have constructed a model for the photodissociation region in Orion. This model matches the observed emission in the fine-structure lines of O I 63 μm and 145 μm , C I 609 μm , and C II 158 μm and the low-lying rotational lines of CO. It also explains the extended H₂ 1-0 S(1) emission, the carbon recombination lines, and the C I 9850 Å line. The emission in the C I 370 μm , Si II 35 μm , and Fe II 26 μm , 35 μm lines is predicted. The model fit implies that a fairly uniform density ($\sim 10^5 \text{ cm}^{-3}$) photodissociation region lies behind the Trapezium stars. The peak temperature in this region is about 500 K, and the intensity of the incident far-ultraviolet (FUV) field is approximately 10^5 times as intense as the ambient interstellar field. This model shows that the observed high brightness temperature of the C I 609 μm line can be explained by emission from the C⁺/C/CO transition region. This difference with previous chemical models is due to a higher gas phase elemental abundance of carbon, to the charge exchange reactions of C⁺ with S and SiO, and to carbon self-shielding.

Subject headings: interstellar: molecules — nebulae: H II regions — nebulae: Orion Nebula

I. INTRODUCTION

In a previous paper we have modeled the chemistry and heating balance in molecular clouds illuminated by an intense far-ultraviolet (FUV) ($6 \text{ eV} < h\nu < 13.6 \text{ eV}$) field (Tielens and Hollenbach 1985, hereafter Paper I). The models assumed an FUV-illuminated plane-parallel slab, so that the calculation was strictly in one dimension, the depth measured in visual extinction A_v into the slab. Microturbulence with a characteristic Doppler width δv_d was assumed to dominate the pressure. Constant hydrogen nucleus density n_0 was assumed, so that the pressure was approximately constant through the slab. The numerical code solved for the equilibrium gas temperature T and the chemistry as a function of A_v . The dominant heating mechanism was found to be the photoelectric ejection of energetic electrons from grains, which transforms up to 0.04 of the incident FUV flux into gas heating. This "heating efficiency" depends on the positive charge of the grains, which in turn depends on a unitless parameter $\gamma \propto G_0 T^{0.5} n_e^{-1}$, with n_e the electron number density and G_0 the FUV flux measured in units of the "Habing field," which is taken to be $1.6 \times 10^{-3} \text{ ergs cm}^{-2} \text{ s}^{-1}$ (Habing 1968). The emergent spectrum from the slab was calculated using an escape probability formalism for the line radiation. The predicted spectra included the fine-structure lines of O I, C I, C II, Fe II, and Si II; the low-lying rotational transitions of CO, and an estimate of the FUV-pumped H₂ 1-0 2 μm vibrational transitions. The spectra depended mainly on the model free parameters n_0 and G_0 . Other model parameters which generally can be derived from other observations or theoretical considerations include the gaseous elemental abundances, δv_d , the ratio of FUV opacity to optical opacity of the grains, the ratio of A_v to column density of hydrogen nuclei, and the emergent far-infrared continuum flux from dust, F_{IR} , which can be calculated from the dust temperature T_0 and the 100 μm dust optical depth $\tau_{100 \mu\text{m}}$.

The results of the models show that the FUV creates a hot (100–1000 K) layer ($A_v < 3$) of atomic gas (H, O, C⁺). This gas is heated by photoelectrons from dust grains and cooled by the

O I 63 μm and C II 158 μm fine-structure lines. Around $A_v \approx 3$ –4 atomic hydrogen and ionized carbon are transformed into H₂ and C or CO, respectively. This warm (50–100 K) gas is also heated by photoelectrons from dust grains but is cooled by molecular rotation lines from CO. Beyond $A_v > 4$ carbon is mainly in the form of CO. The gas is now heated by the infrared continuum from warm ($\sim 75 \text{ K}$) dust grains. Ground state atomic oxygen is radiatively excited by 63 μm continuum photons, followed by collisional de-excitation by H₂ molecules which heats the gas. The gas is also heated by collisions with warm dust grains. Cooling still occurs through the CO rotational lines. Deep in the cloud ($A_v > 10$) atomic oxygen is converted into O₂.

These models are relevant not only to the gas which separates H II regions from their parental molecular cloud but also to neutral shells around planetary nebulae, bright-rimmed clouds, reflection nebulae, and regions surrounding protostars. In this paper we will apply our model to the well-studied Orion photodissociation region. Applications to the other regions are deferred to future publications.

This paper is organized as follows. Section II summarizes the observations of Orion. In § III we analyze the fine-structure observations. The emphasis is on a simple semianalytic model which highlights the essential physics. In this way values for physical parameters of this region such as density, temperature, electron density, and intensity of the incident FUV field are derived. The observations are also compared with our detailed numerical model calculations, and a best fit model is constructed. Section IV summarizes the main results.

II. THE OBSERVATIONS

The Orion complex is the nearest site of recent massive star formation. Most of the observational studies devoted to Orion are reviewed in Glassgold, Huggins, and Schucking (1982). In this section we will summarize these and more recent observations relevant to the understanding of the photodissociation region.

The Orion region has been observed in the O I 63 μm and 145 μm , C II 158 μm , and C I 609 μm fine-structure lines (Melnick, Gull, and Harwit 1979; Furniss *et al.* 1983; Ellis and Werner 1985; Werner *et al.* 1984; Stacey *et al.* 1983; Russell *et al.* 1980; Phillips and Huggins 1981). Table 1 summarizes the observed integrated intensities for these lines, as well as those of the low-lying rotational levels of CO (Phillips *et al.* 1977, 1979). The absolute calibration is uncertain by $\sim 30\%$. The beam sizes of the far-infrared fine-structure line measurements are all about $45''$, except for the C I 609 μm measurement, which has a beam size of $2.6''$. The beam size of the CO observations are about $1''$.

The O I 63 μm emission in Orion has been mapped with $30''$ resolution by Werner *et al.* (1984) and with $1.65''$ by Furniss *et al.* (1983). The emission is extended on a $3'$ scale and shows two distinct peaks. The northern one is centered on the BN/KL region and is probably associated with the wind shock (Werner *et al.* 1984). The southern peak occurs southwest of the Trapezium, slightly south of the radio continuum peak. It is about halfway between the Trapezium and a peak in the far-infrared and submillimeter continuum (Keene, Hildebrand, and Whitcomb 1982; Werner 1983; Drapatz *et al.* 1983) and in the ^{13}CO column density map (Schloerb and Loren 1982). This southern continuum source does not contain any embedded $20 \mu\text{m}$ sources brighter than 60 Jy (less than about 10% of the BN object) and probably represents a column-density enhancement (Keene, Hildebrand, and Whitcomb 1982).

In a northwest-to-southeast scan in the O I 63 μm line across the central region of the Orion Nebula, a third peak shows up at the position of θ^2 Ori A, slightly southeast of the prominent ionization bar (Ellis and Werner 1985). It coincides with a peak in the ^{12}CO as well as in the ^{13}CO brightness temperatures (Schloerb and Loren 1982).

The O I 145 μm and C II 158 μm lines have recently been observed both from the Lear Jet and from the Kuiper Airborne Observatory (KAO) (Stacey *et al.* 1983; Ellis and Werner 1985; Russell *et al.* 1980). The distribution of the O I 145 μm and C II 158 μm intensities in a scan across the Orion Nebula

looks remarkably similar to that of the O I 63 μm intensity. All three show a peak at the Trapezium and at the ionization bar. These far-infrared lines are more extended and spatially uniform than the continuum.

The Orion region has also been scanned with a $2.6''$ beam in the C I 609 μm line (Phillips *et al.* 1980; Phillips and Huggins 1981). The emission in this line is very extended ($\sim 25''$). These observations will be discussed in more detail in § III d.

The molecular gas temperature as derived from ^{12}CO observations has a peak value of about 75 K in the direction of the BN/KL objects and drops to a value of about 10 K at a distance of 1 pc ($\sim 6''$), although the distribution is quite irregular. Distinct patches of higher temperature and column density can be discerned in the CO maps. The ^{12}CO line is expected to become optically thick within the photodissociation region, and therefore the models are constrained to fit these observations.

The carbon radio recombination lines also originate in the photodissociation region. They indicate an electron density of about 10^2 cm^{-3} and a gas temperature in excess of 100 K (Boughton 1978). The red C I lines at 9850 and 8727 Å seem also to be associated with the neutral gas (Münch and Hippelstein 1982). It has been suggested that these lines are emitted during the radiative recombination of C^+ and e (Cesarsky 1982). Recently, Treffers (1983) has observed extended H_2 1–0 S(1) emission in Orion. The intensity of this emission is low, and its association with the ionization bar suggests an origin in the photodissociation region.

In addition to the above observations of fine-structure, H_2 , CO, and carbon lines which we intend to model, there are a number of observations of the H II region, the photodissociation region, and the molecular cloud which can be used to constrain the model parameters.

The Orion Nebula is an H II region which is ionized by the Trapezium stars, in particular θ^1 Ori C, and to a lesser extent θ^2 Ori A. It appears as a blister on the much larger Orion molecular cloud (Zuckerman 1973; Balick, Gammon, and Doherty 1974). The luminosity of the ionizing stars is about

TABLE 1
COMPARISON OF THE MODEL CALCULATIONS WITH OBSERVATIONS TOWARD ORION

Line	Observed at θ^1 Ori C ($\text{ergs cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$)	Standard Model ($\text{ergs cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$)	Model Fit ^a ($\text{ergs cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$)
O I 63 μm	4 to 6 (–2)	7.5(–2)	4.6(–2)
O I 145 μm	3 to 6 (–3)	3.2(–3)	3.5(–3)
C II 158 μm	4 to 7 (–3)	2.6(–3)	3.8(–3)
C I 610 μm	6.7(–6)	3.5(–6)	8.8(–6)
C I 370 μm	...	1.8(–5)	3.8(–5)
Si II 34.8 μm	...	1.6(–3)	1.8(–3)
Fe II 26 μm	...	1.1(–3)	6.0(–4)
Fe II 35.4 μm	...	1.9(–4)	7.0(–5)
CO $J = 1-0$	6.5(–7)	4.1(–7)	4.9(–7)
CO $J = 2-1$	6.0(–6)	4.6(–6)	5.4(–6)
CO $J = 3-2$	1.9(–5)	1.8(–5)	1.9(–5)
CO $J = 4-3$	...	4.1(–5)	4.5(–5)
CO $J = 5-4$	...	7.6(–5)	4.5(–5)
CO $J = 6-5$	...	1.2(–4)	1.1(–4)
CO $J = 7-6$	...	1.5(–4)	1.3(–4)
CO $J = 8-7$	...	1.8(–4)	1.4(–4)
CO $J = 9-8$	...	1.8(–4)	1.3(–4)

NOTE.—Numbers in parentheses: $7.5(-2) = 7.5 \times 10^{-2}$.

^a A possible improved fit with lower than standard dust-to-gas ratio $\delta_d = 0.5$ and a lower gas density $n_0 = 10^5 \text{ cm}^{-3}$ (see text).

$10^5 L_\odot$, and the stars are located at a distance of about 0.1–0.2 pc from the molecular cloud (Werner *et al.* 1976). This leads to an incident FUV flux of about $G_0 \sim 10^5$.

For the elemental abundances, $\mathcal{A}_i$, the values obtained for the heavily depleted line of sight toward ζ Oph are chosen (Morton 1974; de Boer 1981; Jenkins, Jura, and Loewenstein 1983). Note that for the carbon abundance the upper limit has been chosen (see Table 2). This value is equal to the measured carbon abundance in the Orion Nebula (Peimbert 1982) and is therefore probably more appropriate for models of the Orion photodissociation region (the oxygen abundance in Table 2 is also equal to the measured value in the H II region, as expected if the H II plasma consists of ionized molecular cloud material). The derived lower limit to the carbon abundance toward ζ Oph is half the upper limit (Jenkins, Jura, and Loewenstein 1983). As discussed in Paper I, we do not expect that carbon is depleted in the form of grain mantles in the photodissociation region.

The gas density in the central core of the H II region is about 10^4 cm^{-3} . With a gas temperature of about 10^4 K , the H II region gas pressure against the neutral gas is about 10^8 K cm^{-3} . The pressure in the photodissociation region, $P \propto n_0 \delta v_d^2$ (see Paper I), is therefore constrained to be of this order. The gas distribution in the H II region may, however, be quite clumpy, and the external pressure on the photodissociation region may differ somewhat from this value (see Peimbert 1982).

The photodissociation region gas density and pressure are also constrained somewhat by using millimeter observations of the molecular gas which lies deep in the photodissociation region and beyond, in the opaque molecular core. The somewhat dubious premise here is that the pressure and density deep in the cloud reflect the surface pressure and density. These observations reveal a ridge of high-density ($\sim 10^6 \text{ cm}^{-3}$) molecular material running N-S through the BN/KL region

with a gas temperature of perhaps $\sim 50 \text{ K}$. The molecular gas density seems to decrease with a r^{-2} dependence from this region to a value of about 10^3 cm^{-3} at a distance of 1 pc (Westbrook *et al.* 1976; Goldsmith *et al.* 1980; Keene, Hildebrand, and Whitcomb 1982).

Several observations may be used to constrain the turbulent Doppler width δv_d in the photodissociation region. Unfortunately, the line width of the far-infrared fine-structure lines is less than the instrumental resolution ($\sim 30 \text{ km s}^{-1}$) except for the C I 609 μm line, which has an observed FWHM width of 5 km s^{-1} . (We ignore here the small region of shock-excited O I 63 μm emission toward BN/KL which was resolved by Werner *et al.* 1984.) This places an upper limit on δv_d of about 3 km s^{-1} . The carbon recombination lines indicate that the gas in the photodissociation region has $\delta v_d \approx 2.7 \text{ km s}^{-1}$ behind the Trapezium stars (Boughton 1978), although some velocity structure may be present within the fairly large beams (2/6) of these observations (Jaffe and Pankonin 1978). In the calculations we will adopt $\delta v_d = 2.7 \text{ km s}^{-1}$.

Observations which constrain the dust extinction properties are discussed in the next section. However, we note here that the far-infrared continuum observations show a maximum color temperature of the dust of about $T_0 \approx 75 \text{ K}$ and a maximum optical depth at 100 μm ($\tau_{100 \mu\text{m}}$) of about 0.3 (Werner *et al.* 1976).

III. THE MODEL

The parameters of the standard model in Paper I have been chosen to mimic the conditions expected for the photodissociation region directly behind θ^1 Ori C (Table 2). The calculated emergent line intensities for the standard model are compared with the observations in Table 1. Also shown in this table are the results of a possible best fit model. This model has a lower dust-to-gas ratio ($\delta_d = 0.5$ or $A_v/N_H = 2.6 \times 10^{-22} \text{ mag cm}^2$) and a slightly lower gas density (10^5 cm^{-3}) than the standard model ($\delta_d = 1$ or $A_v/N_H = 5.2 \times 10^{-22} \text{ mag cm}^2$, the value appropriate for diffuse clouds, and $n_0 = 2.3 \times 10^5 \text{ cm}^{-3}$). As discussed below, other small changes in the parameters can also produce a good fit. As a result of this change, this model has a somewhat lower temperature throughout the atomic region than the standard model. The maximum temperature, for example, is only 550 K in the best fit model, as opposed to 1100 K in the standard model (cf. Fig. 7 of Paper I). The temperature in the molecular gas is equal to about 50 K in both cases. The behavior of the molecular abundances as a function of visual extinction in this model is quite similar to that in the standard model (cf. Fig. 9 of Paper I). The column density of most species are, however, larger by a factor of 2 because of the increased penetration depth of the FUV.

a) The O I 63 μm Line

The core of our interpretation of the O I 63 μm emission in Orion is that the gas is heated by electrons, photoelectrically ejected from dust grains. As much as a few percent of the energy absorbed by a dust grain can be converted into heat for the gas and ultimately into O I 63 μm emission. The remaining heat will be radiated by the grains in infrared continuum radiation. It is the identification of the heat source of the gas that will allow us to calculate the global physical parameters of the region (e.g., T , n_0 , and G_0) from the observations.

As will be discussed in detail below, the average efficiency of the photoelectric heating can be obtained from the observed

TABLE 2
STANDARD MODEL PARAMETERS^a

Parameter	Standard Model
$n_0 (\text{cm}^{-3})$	2.3(5)
G_0	1.0(5)
$\delta v_d (\text{km s}^{-1})$	2.7
$\mathcal{A}_C$	3.0(–4)
$\mathcal{A}_O$	5.0(–4)
$\mathcal{A}_{Si}$	7.9(–7)
$\mathcal{A}_S$	7.9(–6)
$\mathcal{A}_{Fe}$	2.5(–7)
$\mathcal{A}_{Mg}$	1.3(–6)
$F_{IR} (\text{ergs cm}^{-2} \text{ s}^{-1})$	5.0(2)
$T_0 (\text{K})$	75
$\tau_{100 \mu\text{m}}$	3.0(–1)
δ_d	1.0

NOTE.—Numbers in parentheses: see Table 1.

^a The photodissociation region parameters are defined as follows: n_0 is the hydrogen nucleus density, G_0 is the incident FUV flux in units of the “Habing flux” (see Paper I), δv_d is the turbulent velocity width, $\mathcal{A}_i$ is the fractional abundance of species i , F_{IR} is the integrated far-infrared flux from the slab, T_0 is the dust temperature, $\tau_{100 \mu\text{m}}$ is the dust optical depth at 100 μm , and δ_d is the depletion of dust relative to its abundance in the diffuse interstellar medium, or $\delta_d = (A_v/N_H)/5.2 \times 10^{-22} \text{ mag cm}^2$.

total line and continuum flux. This efficiency depends on the unitless parameter $\gamma = 2.9 \times 10^{-5} G_0 T^{0.5} n_e^{-1}$, with T in K and n_e in cm^{-3} (see Fig. 5 of Paper I). Because the intensity of the incident FUV field can be determined from the dust infrared continuum and the gas temperature from an estimate of the peak intensity of the optically thick O I 63 μm line, a reasonable value of the electron density can be derived. Given the carbon abundance, this determines the gas density in the photodissociation region.

The integrated intensity of all far-infrared fine-structure lines is not much more than the O I 63 μm intensity of 5×10^{-2} ergs $\text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1}$ at the Trapezium. The integrated far-infrared continuum intensity at that position is about 1.2×10^4 ergs $\text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1}$ (Werner *et al.* 1976; Drapatz *et al.* 1983). This yields an efficiency ϵ of the photoelectric effect of about 4×10^{-3} . Actually, this efficiency should be divided by 2. In a semi-infinite plane-parallel layer, a good assumption for the O I 63 μm , the cooling radiation only escapes through one boundary. The far-infrared emission from the dust, however, is optically thin and escapes in two hemispheres. This yields for ϵ about 0.2%. Using Figure 6 from Paper I, we then estimate that $\gamma \approx 0.6$, or $G_0 T^{0.5} n_e^{-1} \approx 2 \times 10^4 \text{ K}^{0.5} \text{ cm}^3$.

The intensity of the incident FUV field G_0 and the gas temperature T can be estimated in the following way prior to detailed model fits. The average interstellar FUV intensity on a cloud is 1.2×10^{-4} ergs $\text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1}$ (Habing 1968). The observed integrated infrared intensity toward the Trapezium (see above) implies that $G_0 = 10^5$ at that position, assuming that FUV is primarily responsible for heating the dust. The integrated intensity of an optically thick line emerging from a homogeneous (constant density and constant excitation temperature T_x) plane-parallel slab is given by

$$I \approx B_\nu(T_x) v(\delta v_d/c) f(\tau), \quad (1)$$

where $B_\nu(T_x)$ is the Planck function at the frequency of the line, δv_d is the turbulent Doppler width, and the factor $f(\tau)$ arises from optical-depth effects, with τ the line-averaged optical depth (Paper I). The function $f(\tau)$ varies logarithmically with τ and is typically of order unity for $\tau \sim 5$. The O I 63 μm line becomes optically thick for hydrogen nucleus column densities $N_H > 2 \times 10^{21} \text{ cm}^{-2}$ or $A_v > 1$. In our detailed model calculations, this line is always optically thick ($\tau \sim 5$) and $f(\tau) \sim 1$. Assuming $\delta v_d = 2.7 \text{ km s}^{-1}$ and using the observed intensity in O I 63 μm , we derive $T_x \approx 265 \text{ K}$.

Using these estimates for G_0 and T_x and assuming that $T = T_x$, we find from the derived value of γ an electron density n_e of 80 cm^{-3} in the photoelectrically heated region. This is in good agreement with detailed models of the carbon recombination lines which originate in the same region (Boughton 1978). If we assume a carbon elemental abundance of 3×10^{-4} as in the models, then the gas density in the photoelectrically heated region is $2.6 \times 10^5 \text{ cm}^{-3}$ in order to explain the ratio of the O I 63 μm to the IR continuum intensity.

The parameters of the models are close to these estimates (Table 2). The O I 63 μm intensity calculated for the standard model is actually somewhat higher than the observed intensity (Table 1). This overestimate is caused by the increase in heating efficiency with increasing depth into the cloud (decreasing FUV field and less positively charged grains), an effect ignored in the simple analysis discussed above. This also explains the higher maximum gas temperature ($\sim 1000 \text{ K}$) in the standard model as compared with the 265 K estimated above. The "best fit" model has a somewhat lower value for the density $n_0 = 10^5$

cm^{-3} , in order to lower the cloud surface heating efficiency somewhat. This model therefore has a maximum temperature of about 500 K, which is closer to the "average" temperature estimated from equation (1).

In a similar way we can derive the physical parameters in the photodissociation region behind the prominent ionization bar southeast of θ^1 Ori C. The integrated O I 63 μm intensity is 4×10^{-2} ergs $\text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1}$, and the total far infrared continuum intensity is 5.3 ergs $\text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1}$ (Werner *et al.* 1976; Ellis and Werner 1985). Therefore, an analogous analysis as described above results in a calculated heating efficiency of 0.4%, an estimated intensity of the FUV field of $G_0 \approx 4.4 \times 10^4$, a gas temperature of 230 K, an electron density of 60 cm^{-3} , and a gas density n_0 of $2 \times 10^5 \text{ cm}^{-3}$. Because of the increase in the efficiency with depth into the slab, the density may have been overestimated by about a factor of 2–3. Despite the lower incident FUV field, the O I 63 μm intensity does not drop appreciably because of the increase in heating efficiency with decreasing G_0 . In other words, the O I 63 μm intensity in the model is more spatially uniform than the far-infrared continuum flux, as is observed.

As an aside, it should be noted that the bar is probably an ionization front seen almost edge-on (Becklin *et al.* 1976; Werner *et al.* 1976; Schloerb and Loren 1982; Keene, Hildebrand, and Whitcomb 1982). However, the almost edge-on geometry does not appreciably alter the calculated O I 63 μm intensities, since the gas temperature is relatively constant on the surface of the cloud where O I 63 μm becomes optically thick to the observer (assuming that the photodissociation slab slightly faces the observer). In order to match calculated intensities with observed intensities, one must assume that the face of the photodissociation region fills the beam.

The derived gas density is about an order of magnitude larger than the lower limit to the density estimated from the extent of the 10 and 20 μm emission associated with the ionization bar (Becklin *et al.* 1976). With our estimated density the physical extent of the atomic region emitting the 10 and 20 μm radiation is only 10^{16} cm . The extent along the line of sight is estimated to be about 0.1 pc (Werner *et al.* 1976). This implies that we view the ionization bar at an angle of 70° rather than 90° . Alternatively, the 10 and 20 μm emission may originate from dust just inside the ionized gas, which has a lower density than the gas in the photodissociation region because of the streaming away of material from the ionization front.

The molecular data provide an insight into the overall density distribution in the molecular cloud behind θ^1 Ori C. These data indicate a density of about 10^5 cm^{-3} at a distance of 3' from the BN/KL region (Goldsmith *et al.* 1980). The region with higher density ($\sim 10^6 \text{ cm}^{-3}$) in the BN/KL region indicated by the molecular data must then be buried deeper ($A_v > 10$) inside the molecular cloud in order that the photodissociation models match the observations. In our view, the formation of the infrared cluster in the BN/KL region has produced a density gradient (r^{-2}) in the surrounding gas because of gravitational collapse. This density distribution is further modified at the front of the molecular cloud by the presence of the H II region around θ Ori C, which produces a roughly circular slab with a projected radius of about 0.45 pc (3') in which the density is quite constant ($\sim 10^5 \text{ cm}^{-3}$). At a distance of about 10', pressure equilibrium between the H II region and the molecular cloud may not yet have been reached, possibly resulting in the propagation of a slow shock into the molecular cloud (Schloerb and Loren 1982). In Figure 1 a

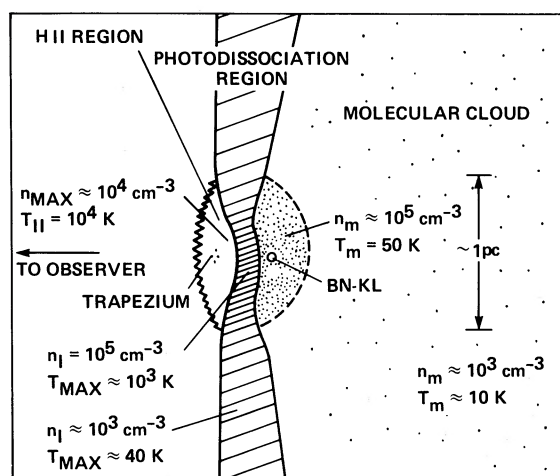


FIG. 1.—Schematic view of Orion region in the model. See text.

schematic view is given of the Orion region in our model. The high-density ($\sim 10^6 \text{ cm}^{-3}$) molecular gas lies approximately within the small circle around BN/KL. Whether the gravitational collapse in the central region has been induced by the pressure of the H II region (Elmegreen and Lada 1977) cannot be ascertained from the infrared fine-structure data.

b) The C II 158 μm Line

The observed intensity of the C II 158 μm line is relatively easy to analyze. Because the density in the photodissociation region is much larger than the critical density, the level populations are given by the Boltzmann distribution function. Because the temperature in the C^+ zone is much larger than the excitation energy (92 K), the level populations are only weakly dependent on T . Thus, for an optically thin line, the C II 158 μm intensity is mainly dependent on the total C^+ column density. For an optically thick line the C II 158 μm intensity is given by the Planck function at the kinetic temperature (cf. eq. [1]). In Paper I a simple analytical model was used to derive the intensity of the C II 158 μm line as a function of the total C^+ column density and the temperature (see Fig. 5 in Paper I). For our best estimate of turbulent Doppler width $\delta v_D = 2.7 \text{ km s}^{-1}$ in the photodissociation region, the observed C II 158 μm intensity implies a temperature of at least $\sim 250 \text{ K}$, in good agreement with values estimated from the O I 63 μm line. The observations also imply a total C^+ column density of about $5 \times 10^{18} \text{ cm}^{-2}$.

The column density of C^+ is mainly determined by the penetration of FUV photons into the slab. Typically, the C^+ region extends to a grain optical depth at $\lambda \sim 1000 \text{ \AA}$ of $\tau \approx 10$. Assuming “normal” dust properties ($\tau/A_v = 2.6$ and $A_v/N_H = 5.2 \times 10^{-22} \text{ mag cm}^2$), $N_H = 6.8 \times 10^{21} \text{ cm}^{-2}$ is derived. Taking the above C^+ column densities at face value, this implies a carbon abundance of 7.5×10^{-4} , larger than the cosmic abundance value of 4×10^{-4} (Allen 1973) or the carbon abundance in the Orion H II region (Peimbert 1982).

Rather than a superabundance of carbon, the C^+ column density can be raised by two more probable causes. First, the dust in molecular clouds may not have “normal” properties. The extinction curve of the dust along the line of sight to the Trapezium stars differs considerably from the average interstellar extinction curve. The ratio of total to selective extinction, $R = A_v/E_{B-V}$, is larger, and the 2200 $\text{\AA}$ bump and the

ratio of FUV opacity to A_v is smaller ($\tau/A_v \sim 1$), than in the diffuse interstellar medium (Johnson 1968; Bless and Savage 1972; Panek 1983). These differences, in dust which is presumably leftover molecular cloud material, seem to point to a larger mean size of the grains rather than the absence of small grains. The larger grain size may be caused by grain processing in the molecular cloud—for example, by coagulation of large and small grains. If the photodissociation region dust has a similar low FUV opacity, this would increase the extent of the C^+ region. Second, the large C^+ column density may be due to geometrical effects. Viewing the slab at an angle with the normal increases the observed column density, without increasing the extent of the C^+ region. In order to explain the intensities observed in Orion, the slab has to be tilted by about 60° .

These simple analytical considerations agree well with our extensive numerical calculations. The C II 158 μm intensity in the standard model falls short of the observations by about a factor of 2 (Table 2). The C II 158 μm line intensity in the possible best fit model, which has a lower dust-to-gas ratio than the standard model, is in much better agreement with the observations. It should be emphasized, however, that an equally good fit to the observations can be obtained by tilting the slab or by choosing a normal gas-to-dust ratio combined with a gray FUV dust opacity ($\tau/A_v = 1$). At present we cannot rule out any of these possibilities, and it is possible that all are operating. We do note, however, that the observed variations in the C II 158 μm line intensities in Orion are probably due to variations in geometry rather than to dust properties.

c) The O I 145 μm Line

The intensity in the O I 145 μm line is a complicated function of density, temperature, and optical depth in both the O I 63 μm and 145 μm lines. It is therefore difficult to interpret the observed O I 145 μm intensities in terms of a simple analytical model. Instead the full-scale numerical model calculation is needed. The intensity of the O I 145 μm line in the standard model is in reasonable agreement with the observations (Table 1). The ratio of the O I 145 μm to the O I 63 μm line may be somewhat small compared with the observations, although it should be kept in mind that the observed intensities are uncertain by about $\pm 30\%$. We note that if this observed ratio is as high as 0.1, then both transitions are optically thick (cf. the curve for $n_0 = 10^5 \text{ cm}^{-3}$ in Fig. 2 of Paper I). In the standard model the O I 63 μm line is optically thick but the O I 145 μm line is still optically thin. Increasing the O column density, as in the possible improved model fit (Table 2), does indeed increase the O I 145 μm /O I 63 μm line ratio. In this model the O I 145 μm line does reach an optical depth of about unity in the warm gas.

d) Origin of the C I 609 μm and Millimeter CO Emission

The C I fine-structure lines as well as the CO rotational levels originate in the warm (50–200 K) molecular gas deep in the photodissociation region. The calculated intensity of the C I 609 μm line in the standard model is about a factor of 2 lower than the observed intensity in Orion (Table 2). Because this line is optically thin ($\tau \sim 0.5$) in our model, this discrepancy is due to a C I column density which is somewhat too small. As discussed in relation to the C II 158 μm line, this may be caused by geometrical effects or to a difference in the dust properties. This discrepancy may also be attributed to a higher C/O ratio, which tends to shift the C/CO equilibrium toward C

(see Paper I). In view of the C II 158 μm results, however, the first two explanations are preferred. As exemplified by the results of the possible best fit model in Table 2, a good fit to the observations can be easily obtained.

The calculated intensities for the CO 1-0, 2-1, and 3-2 transitions are in reasonable agreement with the observations (brightness temperatures ≈ 75 K). It should be noted that the brightness temperature of the CO rotational transitions in the model increases with increasing J for low values of J . This is caused by the increase in Einstein A -value with J which makes the higher J -levels optically thick closer to the (warmer) surface. This effect may have been observed in the $J = 6-5$ transition, which is measured to have a brightness temperature of 120 K (Buhl *et al.* 1982).

The column density of neutral carbon in the photodissociation region model is about 10 times larger than previous calculations indicated (Glassgold and Langer 1975; Langer 1976, 1977; de Jong, Dalgarno, and Boland 1980). As discussed in Paper I, this is partly caused by improvements in the chemical modeling: we include charge exchange reactions between C^+ and S and SiO and self-shielding of neutral carbon. The rest of the difference is due to the choice of a higher elemental abundance of carbon in the gas phase than previous studies have assumed (cf. Paper I). This is consistent with recent determinations of the carbon depletion in the diffuse interstellar medium (Jenkins, Jura, and Loewenstein 1983). It should also be emphasized that a high carbon abundance is also needed to explain the C II 158 μm observations.

The low column densities of atomic carbon predicted by previous gas-phase chemistry models failed to provide the observed C I 609 μm intensities. The failure of these chemical equilibrium models has led to several speculations in the literature on the origin of the high atomic carbon column densities inferred from observation.

It has been suggested that the high atomic carbon column densities are caused by the lack of chemical equilibrium in molecular clouds. The chemical equilibrium time scale inside a dense cloud, where the chemistry is dominated by cosmic-ray ionization, is about 10^6 yr. Starting with an atomic cloud (i.e., all carbon in C^+), large abundances of C I are present for $t < 10^6$ yr in time-dependent models (Iglesias 1977; Gerola and Glassgold 1978; Graedel, Langer, and Frerking 1982; Tarafdar *et al.* 1985). However, this young implied age for molecular clouds is in conflict with other observations which suggest ages of 10^7 – 10^8 yr (cf. Bash and Peters 1976; Bash, Green, and Peters 1977). In particular, the presence of several subassociations suggest a lifetime of the Orion molecular cloud larger than 10^7 yr (Elmegreen and Lada 1977).

The nonequilibrium of the molecular cloud might be caused by turbulence. Boland and de Jong (1982) suggest that churning of atomic material from the surface to the center of the cloud leads to large amounts of C I, because of the photodissociation of CO at the surface and the long time scale for re-formation of CO in the center. However, it should be noted that an FUV radiation field capable of photodissociating CO is also capable of photoionizing C to C^+ . Furthermore, the chemical equilibrium time scale in the presence of an FUV field is much shorter than the 10^6 yr quoted above for the interiors of dense clouds (a fact noted by Boland and de Jong 1982). As estimated in Paper I, the chemical equilibrium time scale in photodissociation regions is only 10^4 yr, much shorter than the turbulent circulation time scale. Consequently, photodissociation regions are expected to be in chemical equi-

librium, and the carbon brought to the interior by turbulence should be mainly in the form of CO. Indeed, detailed model calculations of Boland and de Jong (1984) show atomic carbon in the photodissociation region of a dark molecular cloud, but the center is entirely CO.

Our photodissociation model for the C I 609 μm emission places the C I in chemical equilibrium at the cloud surface ($A_v < 10$). The emission is barely optically thin ($\tau \sim 1$) and originates in a warm (50 K) molecular gas. Our models predict about 10 times greater atomic carbon column densities than previous theoretical models and match the observationally inferred (optically thin) lower limits on C column densities. In the remainder of this section we argue for these lower limits on the observed C I column densities rather than the higher values favored by Phillips and Huggins (1981). They propose that C I 609 μm is very optically thick, based on a curve-of-growth analysis.

The emission from the C I 609 μm line is very extended ($\sim 25'$), and its brightness temperature is about equal to that of the $^{12}\text{CO } J = 1-0$ line in the outer regions which lie $> 10'$ from θ^1 Ori C. In the central region the brightness temperature of the C I 609 μm line is only about one-third that of the $^{12}\text{CO } J = 1-0$ line. In our model this behavior of the C I 609 μm line is caused by an increase in optical depth with distance from θ^1 Ori C. In the central region the C I 609 μm line has an optical depth < 1 in our model; in the outer regions this line is optically thick. This difference in optical depth is caused not by an increase in the C I column density but rather by a decreased turbulent Doppler width. The C I 609 μm line width decreases from about 5 km s^{-1} in the central region to about 2 km s^{-1} 25' away (T. G. Phillips 1984, private communication). Therefore, in the outer regions where they are both optically thick, the $^{12}\text{CO } J = 1-0$ and C I 609 μm lines measure essentially the kinetic gas temperature and are insensitive to n_0 and G_0 .

Phillips and Huggins (1981) have suggested that the C I 609 μm line is very optically thick even in the center of the OMC-1 region. This conclusion was reached on the basis of a comparison of its line width to that of the ^{12}CO lines and its isotopes. We feel that such a curve-of-growth analysis of this complex region may be misleading. The presence of temperature, density, and turbulent or systematic velocity gradients can lead to erroneous conclusions based on line-width comparisons. For example, the C I 609 μm and $^{13}\text{CO } J = 2-1$ lines are both optically thick in the central region ($\tau_{\text{C I}} \sim 25$), according to the analysis of Phillips and Huggins (1981), yet their brightness temperatures are much less than the kinetic gas temperature derived from the $J = 1-0$ ^{12}CO line ($\tau \sim 80$).

In our opinion, this difficulty stems from the presence of two regions superposed on the sky: (1) the warm OMC-1 core around BN/KL, including the warm photodissociation region behind the Trapezium, and (2) the cold, lower density molecular gas behind BN/KL (see Fig. 1). These regions have quite different densities, temperatures, column densities, and intrinsic line widths. The H_2CO and CO observations, for example, show complex velocity structure (Bastien *et al.* 1981; Schloerb and Loren 1982). The H_2CO line width (FWHM) is only $1-2 \text{ km s}^{-1}$, which may reflect quiescent conditions much deeper into the Orion molecular cloud, far beyond the photodissociation region. Therefore, we believe that the colder quiescent region has an intrinsic line width (FWHM) of about $1-2 \text{ km s}^{-1}$, while the region around and in front of BN/KL has a line width (FWHM) of about 5 km s^{-1} (or $\delta v_d = 2.7 \text{ km s}^{-1}$), as inferred from the carbon recombination lines. The CO

lines with high opacity (e.g., $^{12}\text{CO } J = 2-1$ and $J = 1-0$ and $^{13}\text{CO } J = 2-1$) originate mainly in the warm turbulent region. Those of lower opacity (e.g., $^{13}\text{CO } J = 1-0$ and $^{12}\text{C}^{18}\text{O } J = 1-0$ and $J = 2-1$) have a significant contribution from the cold quiescent molecular cloud. Thus, in our model, the increased width of a line does not imply greatly increased column densities, as in a curve-of-growth analysis, but rather is indicative of variations in the velocity field with depth into the cloud. It is apparent from this discussion, however, that high spectral resolution observations of the far-infrared fine-structure lines are important. The $\text{C II } 158 \mu\text{m}$ and $\text{O I } 145 \mu\text{m}$ lines are ideally suited for determination of δv_d , since they are generally optically thin and both arise in the photodissociation region.

Further discussion of this point must await a more thorough radiative transfer calculation than ours. However, we wish to emphasize that the ^{12}CO and $\text{C I } 609 \mu\text{m}$ line intensities calculated for the photodissociation region are in reasonable agreement with the observations (Table 2). Although we have not included ^{13}CO in our chemical calculations, the following simple argument shows that the ^{13}CO line intensities can also be obtained in our model. Assuming that the $^{12}\text{CO}/^{13}\text{CO}$ ratio is cosmic (90) and that the kinetic temperature is constant (70 K) over the emitting region, we obtain brightness temperatures from the photodissociation region of 20 and 7 K for the $^{13}\text{CO } J = 2-1$ and $J = 1-0$ lines, respectively. The value for the 2-1 line is in reasonable agreement with the observations. Phillips *et al.* (1979) observed a width corresponding to $\delta v_d \approx 2.9 \text{ km s}^{-1}$ for the $^{13}\text{CO } J = 2-1$ line, consistent with our assumed width for the photodissociation region. The $J = 1-0$ line, however, has a model peak brightness temperature which is about a factor of 2 lower than the observation. Because of the low optical depth of this line in the photodissociation region, it may have a substantial contribution from the cold quiescent molecular cloud. This may explain the observed width of the $^{13}\text{CO } J = 1-0$ line, $\delta v_d \approx 2.3 \text{ km s}^{-1}$, which is somewhat narrower than lines which originate purely in the photodissociation region.

From this discussion it may appear that measuring the $\text{C I } 370 \mu\text{m}$ line in OMC-1 will be helpful in distinguishing whether atomic carbon is mainly at the cloud surface with line opacities of order unity or whether the carbon exists throughout the cloud with correspondingly large opacities. The latter model predicts equal brightness temperatures in the lines. In our model, however, the brightness temperatures depend on the optical depths in the lines and, *a priori*, these might be expected to differ. The optical depths in the 370 and 609 μm lines are, however, quite similar, so that the brightness temperatures of the two lines are similar even in the optically thin case (i.e., both models predict about the same ratio of intensities $\text{C I } 370 \mu\text{m}/\text{C I } 609 \mu\text{m}$). In view of the uncertainty in the absolute calibration of submillimeter lines, it appears that measuring the $\text{C I } 370 \mu\text{m}$ intensity will not settle this question.

e) $\text{C I } 9850 \text{ \AA}$

The observed intensity of the 9850 $\text{\AA}$ C I line is somewhat uncertain because of an unknown reddening correction. In our model this line originates within the region $0 < A_v < 3$. Assuming that the wavelength dependence of the dust is represented by van de Hulst curve No. 15, this corresponds to an optical depth at 9850 $\text{\AA}$ of about 0.5. Correcting the observed intensity (Münch and Hippelein 1982) by this amount of

extinction results in an emitted intensity of $1.1 \times 10^{-4} \text{ ergs cm}^{-3} \text{ s}^{-1} \text{ sr}^{-1}$. It should, however, be noted that this correction for extinction is uncertain in the photodissociation region. The calculated intensity in the standard model is about $9 \times 10^{-5} \text{ ergs cm}^{-3} \text{ s}^{-1} \text{ sr}^{-1}$, in good agreement with this value, lending further support to the suggestion that this line originates from carbon recombination in the photodissociation region (Cesarsky 1982; Münch and Hippelein 1982).

f) $\text{H}_2 1-0 S(1)$

The extended emission in the $\text{H}_2 1-0 S(1)$ line, which we attribute to FUV-pumped H_2 molecules in the photodissociation region, has an average intensity of about $2 \times 10^{-4} \text{ ergs cm}^{-3} \text{ s}^{-1} \text{ sr}^{-1}$ (Treffers 1983). This implies a column density of H_2 in the $v = 1, J = 3$ level of about $7.5 \times 10^{15} \text{ cm}^{-2}$. Assuming that the H_2 molecules in the $v = 1$ state are distributed over the rotational levels according to the kinetic temperature of the gas ($\sim 500 \text{ K}$), we compute a total column density of H_2 molecules in $v = 1$ of about $2 \times 10^{16} \text{ cm}^{-2}$. According to Shull's (1978) accurate calculations, the column density of H_2 in all $v \geq 1$ states is about 2.5 times that in the $v = 1$ state. This leads to a total "observed" column density of H_2^* of about $5 \times 10^{16} \text{ cm}^{-2}$, which is in good agreement with the predicted standard model value of $4.4 \times 10^{16} \text{ cm}^{-2}$ (see Paper I).

It should perhaps be noted that the possible best fit model of Table 1 predicts a H_2^* column density which is about equal to that in the standard model. In both models H_2^* peaks and then rapidly falls off at $A_v > 2$, where dust and H_2 self-shielding both attenuate FUV pumping. The best fit model has a ratio A_v/N_H one-half that of the standard model, so that the extent of the H_2^* region is twice that of the standard model. However, this is offset by the abundance of H_2^* , which is one-half that of the standard model because of the factor of 2 decrease in H_2 formation on grains. If, instead, we had changed the FUV opacity of the dust to $\tau = A_v$, then again we would have increased the extent of the H_2^* region (or the penetration of the FUV) but without lowering the H_2 formation rate. Thus, in this case, the H_2^* column density would have increased by a factor of 2, as H_2 competed more effectively with the dust for FUV photons.

Finally, we note that the detection of extended emission from vibrationally excited H_2^* in Orion implies that H_2 can be formed with high efficiency on warm dust grains. It is difficult to make this more quantitative because of the possible presence of multiple dust components and different sizes, resulting in different grain temperatures in a given region (cf. Paper I).

IV. SUMMARY

We have applied our theoretical model, which calculates the energy balance and chemical equilibrium of dense molecular clouds illuminated by high FUV fields, to the photodissociation region in Orion. Our model can explain the observed intensities in the $\text{O I } 63 \mu\text{m}$ and $145 \mu\text{m}$, $\text{C II } 158 \mu\text{m}$, $\text{C I } 609 \mu\text{m}$, and the low-lying rotational transitions of CO. It also explains the extended vibrationally excited H_2 emission, the carbon radio recombination lines, and the far-red $\text{C I } 9850 \text{ \AA}$ emission.

We have illustrated how the $\text{O I } 63 \mu\text{m}$, $\text{C II } 158 \mu\text{m}$, and IR continuum maps can be used, without the detailed thermochemical modeling described in Paper I, to probe the physical state of the photodissociation region. The ratio of the sum of

the integrated line strengths to the IR continuum gives an efficiency of gas to grain heating by the FUV incident on the cloud. This efficiency depends on the grain charge, which in turn depends on n_e , G_0 , and T . The FUV flux G_0 can be obtained from the IR continuum; the gas temperature T can be estimated from the integrated O I 63 μm or C II 158 μm intensity together with an assumed line width. The measured efficiency, together with G_0 and T , provides the electron density if the photoelectric heating mechanism is valid. The abundance of carbon, which is constrained by the C I and C II fine-structure line strengths, then can be used to derive n_0 from n_e .

A detailed thermochemical one-dimensional model fit to the observed emission from the photodissociation region associated with the Trapezium in Orion is obtained with a gas density of $n_0 = 10^5 \text{ cm}^{-3}$ and an incident FUV flux 10^5 times the average interstellar value. The FUV flux implies that θ^1 Ori C lies about 0.2 pc from the molecular cloud. The gas density in the photodissociation region is relatively constant to a distance $< 3'$ from θ^1 Ori C. The dense ($\sim 10^6 \text{ cm}^{-3}$) molecular ridge at BN/KL must lie buried deeper ($A_v > 10$) in the cloud. The match to the extended H₂ 1–0 S(1) emission implies that H₂ formation occurs efficiently on grains with temperatures $T_{\text{gr}} \sim 75 \text{ K}$.

The large extent of the O I 63 μm emission is a natural consequence of the photoelectric heating mechanism. When the FUV flux drops, the positive grain charge decreases, resulting

in a larger heating efficiency per FUV photon. As a consequence, the intensity of the O I 63 μm line, the dominant cooling line of the gas, does not drop as fast as the far-infrared continuum with distance from the exciting star.

Most of the C I 609 μm emission and the ¹²CO 1–0 emission observed toward OMC-1 originates in the photodissociation region. The ¹²CO 1–0 transition becomes optically thick at $A_v \sim 4$, and the peak brightness temperature measures the gas temperature at this depth. The C I 609 μm emission is marginally optically thin, so that its peak brightness temperature ($\sim 20 \text{ K}$) lies below the gas temperature. We argue that it is not necessary to invoke nonequilibrium chemistry, and therefore extremely young ages $< 10^6$ years, in order to explain C I abundances observed toward molecular clouds. To explain the CO millimeter emission, we propose a two-component model with a warm foreground turbulent region (which includes the photodissociation region) and a cold background quiescent region. Some of the CO lines (e.g., ¹²CO 1–0 and 2–1 and ¹³CO 2–1) come from the turbulent region; others (e.g., ¹³CO 1–0 and ¹²C¹⁸O and 2–1) come from the cold molecular cloud.

We have benefited from discussions with R. Genzel, D. Watson, and M. Werner. We would like to thank the Astronomy/Relativity Branch at NASA Headquarters for their support under NASA RTOP 188-41-53.

REFERENCES

- Allen, C. W. 1973, *Astrophysical Quantities* (London: Athlone).
 Balick, B., Gammon, R. H., and Doherty, L. H. 1974, *Ap. J.*, **188**, 45.
 Bash, F. N., Green, E., and Peters, W. L. 1977, *Ap. J.*, **217**, 464.
 Bash, F. N., and Peters, W. L. 1976, *Ap. J.*, **205**, 786.
 Bastien, P., Bieging, J., Henkel, C., Martin, R. N., Pauls, T., Walmsley, C. M., Wilson, T. L., and Ziurys, L. M. 1981, *Astr. Ap.*, **98**, L4.
 Becklin, E. E., Beckwith, S., Gatley, I., Matthews, K., Neugebauer, G., Sarazin, C., and Werner, M. W. 1976, *Ap. J.*, **207**, 770.
 Bless, R. C., and Savage, B. D. 1972, *Ap. J.*, **171**, 293.
 Boland, W., and de Jong, T. 1982, *Ap. J.*, **261**, 110.
 ———. 1984, *Astr. Ap.*, **134**, 87.
 Boughton, W. L. 1978, *Ap. J.*, **222**, 517.
 Buhl, D., Chin, G., Koepf, G. A., Peck, D. D., and Fetterman, H. R. 1982, in *Submillimeter Wave Astronomy*, ed. J. E. Beckman and J. P. Phillips (Cambridge: Cambridge University Press).
 Cesarsky, D. A. 1982, *Astr. Ap.*, **113**, L7.
 de Boer, K. S. 1981, *Ap. J.*, **244**, 848.
 de Jong, T., Dalgarno, A., and Boland, W. 1980, *Astr. Ap.*, **91**, 68.
 Drapatz, S., Haser, L., Hofmann, R., Oda, N., and Iyengar, K. V. K. 1983, *Astr. Ap.*, **128**, 207.
 Ellis, H. B., and Werner, M. W. 1985, in preparation.
 Elmegreen, B. G., and Lada, C. J. 1977, *Ap. J.*, **214**, 725.
 Furniss, I., Jennings, R. E., King, K. J., Lightfoot, F. G., Emery, R. J., Naylor, D. A., and Fitton, B. 1983, *M.N.R.A.S.*, **202**, 859.
 Gerola, H., and Glassgold, A. E. 1978, *Ap. J. Suppl.*, **37**, 1.
 Glassgold, A. E., Huggins, P. J., and Schucking, E. L., eds. 1982, *Symposium on the Orion Nebula to Honor Henry Draper* (Ann. NY Acad. Sci., Vol. **395**).
 Glassgold, A. E., and Langer, W. D. 1975, *Ap. J.*, **197**, 347.
 Goldsmith, P. F., Langer, W. D., Schloerb, F. P., and Scoville, N. Z. 1980, *Ap. J.*, **240**, 524.
 Graedel, T. E., Langer, W. D., and Frerking, M. A. 1982, *Ap. J. Suppl.*, **48**, 321.
 Habing, H. J. 1968, *Bull. Astr. Inst. Netherlands*, **19**, 421.
 Iglesias, E. R. 1977, *Ap. J.*, **218**, 697.
 Jaffe, D. T., and Pankonin, V. 1978, *Ap. J.*, **226**, 869.
 Jenkins, E. B., Jura, M., and Loewenstein, M. 1983, *Ap. J.*, **270**, 88.
 Johnson, H. L. 1968, in *Stars and Stellar Systems*, Vol. 7, *Nebulae and Interstellar Matter*, ed. B. M. Middlehurst and L. H. Aller (Chicago: University of Chicago Press), p. 167.
 Keene, J., Hildebrand, R. H., and Whitcomb, S. E. 1982, *Ap. J. (Letters)*, **252**, L11.
 Langer, W. D. 1976, *Ap. J.*, **206**, 699.
 ———. 1977, *Ap. J. (Letters)*, **212**, L39.
 Melnick, G., Gull, G. E., and Harwit, M. 1979, *Ap. J. (Letters)*, **227**, L29.
 Morton, D. C. 1974, *Ap. J. (Letters)*, **143**, L35.
 Münch, G., and Hippelein, H. 1982, in *Symposium on the Orion Nebula to Honor Henry Draper*, ed. A. E. Glassgold, P. J. Huggins, and E. L. Schucking (Ann. NY Acad. Sci., Vol. **395**), p. 170.
 Panek, R. J. 1983, *Ap. J.*, **270**, 169.
 Peimbert, M. 1982, in *Symposium on the Orion Nebula to Honor Henry Draper*, ed. A. E. Glassgold, P. J. Huggins, and E. L. Schucking (Ann. NY Acad. Sci., Vol. **395**), p. 24.
 Phillips, T. G., and Huggins, P. J. 1981, *Ap. J.*, **251**, 553.
 Phillips, T. G., Huggins, P. J., Kuiper, T. B. H., and Miller, R. E. 1980, *Ap. J. (Letters)*, **238**, L103.
 Phillips, T. G., Huggins, P. J., Neugebauer, G., and Werner, M. W. 1977, *Ap. J. (Letters)*, **217**, L161.
 Phillips, T. G., Huggins, P. J., Wannier, P. G., and Scoville, N. Z. 1979, *Ap. J.*, **231**, 720.
 Russell, R. W., Melnick, G., Gull, G. E., and Harwit, M. 1980, *Ap. J. (Letters)*, **240**, L99.
 Schloerb, F. P., and Loren, R. B. 1982, in *Symposium on the Orion Nebula to Honor Henry Draper*, ed. A. E. Glassgold, P. J. Huggins, and E. L. Schucking (Ann. NY Acad. Sci., Vol. **395**), p. 32.
 Shull, J. M. 1978, *Ap. J.*, **219**, 877.
 Stacey, G. J., Smyers, S. D., Kurtz, N. T., and Harwit, M. 1983, *Ap. J. (Letters)*, **265**, L7.
 Tarafdar, S. P., Prasad, S. S., Huntress, W. T., Jr., Villere, K. R., and Black, D. C. 1985, *Ap. J.*, **289**, 220.
 Tielens, A. G. G. M., and Hollenbach, D. 1985, *Ap. J.*, **291**, 722 (Paper I).
 ———. 1984, *Icarus*, in press.
 Treffers, R. 1983, private communication.
 Werner, M. W. 1983, private communication.
 Werner, M. W., Crawford, M. K., Genzel, R., Hollenbach, D. J., Townes, C. H., and Watson, D. M. 1984, *Ap. J. (Letters)*, **282**, L81.
 Werner, M. W., Gatley, I., Harper, D. A., Becklin, E. E., Loewenstein, R. F., Telesco, C. M., and Thronson, H. A. 1976, *Ap. J.*, **204**, 420.
 Westbrook, W. E., Werner, M. W., Elias, J. H., Gezari, D. Y., Hauser, M. G., Lo, K. Y., and Neugebauer, G. 1976, *Ap. J.*, **209**, 94.
 Zuckerman, B. 1973, *Ap. J.*, **183**, 863.